

Deliverable 1: User Stories Document

Project: MedSchedule Healthcare Scheduling System | Date: August 2026

1. Executive Summary & Project Context

MedSchedule is an enterprise-grade medical scheduling and clinical workflow management platform designed to automate patient appointment booking, multi-role staff shift management, and patient record integration. The system eliminates double-booking, accommodates specialized clinical requirements (e.g., manual approval for Surgeons and Anesthetists), and dispatches automated SMS/email reminders.

2. System User Roles & Persona Matrix

Role Group	Specific Roles	Primary Responsibilities in System
Patients	Patient	Registers profile, searches available slots, books appointments, receives SMS/email alerts.
Clinical Staff	Doctor, Nurse, Pharmacist, Lab Scientist, Radiologist, Physiotherapist, Dentist, Dietitian	Sets weekly availability & break times, conducts clinical consultations, reviews patient medical records.
Specialist Staff	Surgeon, Anesthetist	Requires manual approval step before confirming pending patient bookings due to surgical prep.
Administrative	Receptionist, Admin Staff, Manager, IT Staff	Books on behalf of walk-ins, manages user accounts, monitors system analytics and schedules.

3. Epic Specifications & Agile User Stories

US-1.1: Patient Account Registration & Medical Profile Setup	Priority: High Points: 5
User Story: As a <i>Patient</i> , I want to register an account with my phone number (+234 format), email, blood group, genotype, and emergency contact details, so that clinical staff can view my health context before appointments and contact emergency relations if needed.	
Acceptance Criteria: Given a new patient, when they submit valid registration info with a unique +234 phone number, then a User and PatientProfile model instance are created automatically.	
US-2.1: Weekly Shift & Break Time Configuration	Priority: High Points: 8
User Story: As a <i>Doctor / Clinical Staff</i> , I want to define my working days, start/end shift hours, slot duration (e.g. 30 mins), and recurring break times (e.g. Lunch 1pm-2pm), so that the system can automatically generate available consultation slots while blocking break hours.	
Acceptance Criteria: Given a doctor's active StaffAvailability and BreakTime records, when the slot generator script executes, then 30 days of AppointmentSlot entities are created excluding break intervals.	
US-3.1: Real-Time Slot Search and Booking	Priority: High Points: 5
User Story: As a <i>Patient / Receptionist</i> , I want to search available slots by doctor, clinical service type, and date, so that I can instantly select and reserve an open appointment time without double-booking.	

Acceptance Criteria: Given available slots for a selected date, when a patient submits a booking request, then the slot status updates to 'booked' and an Appointment instance is generated.

US-3.2: Specialist Manual Approval Workflow

Priority: High | Points: 5

User Story: As a *Surgeon / Anesthetist*, I want to review pending appointment requests before final confirmation, so that I can verify theater availability and patient pre-op requirements.

Acceptance Criteria: Given a slot booked for a Surgeon or Anesthetist, when the patient books, then the appointment status remains 'pending' until explicitly approved by the specialist.

US-4.1: Multi-Channel Booking Confirmation & Reminders

Priority: Medium | Points: 3

User Story: As a *Patient & Staff Member*, I want to receive automated SMS, email, and in-app notifications upon booking creation or status updates, so that both parties stay informed and appointment no-show rates are minimized.

Acceptance Criteria: Given an appointment status transition, when the backend triggers `send_notification()`, then a Notification record is saved and SMS/Email dispatches are executed according to user preferences.

US-5.1: Operational Overview & Analytics

Priority: Medium | Points: 5

User Story: As a *Hospital Manager / Administrator*, I want to view system-wide metrics including total bookings, completed consultations, active staff, and upcoming meetings, so that I can monitor hospital throughput and schedule staff meetings efficiently.

Acceptance Criteria: Given an authenticated Manager, when navigating to the dashboard, then real-time aggregated metrics and schedule summaries are rendered.